Errata for B. C. Carlson, *Special Functions of Applied Mathematics* (1977)

Bastiaan J. Braams

2026-10-26

Abstract

This report lists printed statements in Carlson's book [1] that were found to be false, or to need a qualification, while the book was formalized in the Lean 4 project `lean-codes`. Each entry gives the statement as printed, a counterexample, a correction, and the Lean declarations that record it. All Lean results cited depend only on the standard axioms (`propext`, `Classical.choice`, `Quot.sound`).

1 Summary

Location	Kind	Lean
Theorem 6.6-2	False as printed (lower bound)	`carlsonRPolynomial_coincident_counterexample`
Exercise 6.9-17, last two formulas	False on half of the stated domain	`carlsonRC_cot_sq_csc_sq_complex`, `carlsonRC_coth_sq_csch_sq_complex`
Theorem 7.4-2	Definition needs a qualification	`carlson_polynomial_two_saddle_ratio_counterexample`
Table 9.3-1, row $2b = (1,3,3)$	Wrong sign of C	`carlsonR_table_9_3_1_row4`
Exercise 9.7-2	False as stated (missing branch condition)	`carlsonRF_add_carlsonRF_shift`
Exercise 9.8-6	False as stated (missing sign condition)	`integral_quartic_eq_integral_cubic_of_nonneg`

2 Entries

2.1 Theorem 6.6-2 (growth of R-polynomials)

As printed For $z_i \neq 0$ and $c = \sum b_i$ not a nonpositive integer,

$$\limsup_{n\to\infty} \|R_n(b,z)\|^{1/n} = \max_i \|z_i\|.$$

Counterexample $b = (\frac{1}{2}, -\frac{1}{2}, 1)$, $z = (2,2,1)$. All printed hypotheses hold, but the generating function (6.6-1) is

$$(1-2s)^{-1/2}(1-2s)^{1/2}(1-s)^{-1} = (1-s)^{-1},$$

and since $(c)_n/n! = 1$, $R_n(b,z) = 1$ for every n. The limsup is 1, not 2.

Cause The proof asserts that the singularity of each factor $(1-tz_i)^{-b_i}$ survives in the product; parameters at a repeated node can cancel.

Correction The upper bound $\limsup \|R_n(b,z)\|^{1/n} \le \max_i \|z_i\|$ holds as printed. For the lower bound, either require pairwise distinct nodes with the parameter at a maximal node not a nonpositive integer, or first aggregate equal nodes and require that the summed parameter at a maximal node is not a nonpositive integer.

Lean `Carlson.carlsonRPolynomial_coincident_counterexample`, `Carlson.eventually_norm_carlsonRPolynomial_le`, `Carlson.frequently_le_norm_carlsonRPolynomial` (`Carlson.RPolynomial.Growth`), `Carlson.frequently_le_norm_carlsonRPolynomial_aggregate` (`Carlson.Aggregation`).

2.2 Exercise 6.9-17 (last two formulas)

As stated For complex $\theta, \varphi \neq 0$ with $|\operatorname{Re}\theta| < \pi/2$ and $|\operatorname{Im}\varphi| < \pi/2$,

$$\theta = R_C(\cot^2\theta, \csc^2\theta), \qquad \varphi = R_C(\coth^2\varphi, \operatorname{csch}^2\varphi).$$

Counterexample The right sides are even functions of the angle, the left sides are odd. For $\theta = -\pi/4$ the right side is $R_C(1,2) = \pi/4$ (the exercise itself states $4R_C(1,2) = \pi$), not $-\pi/4$; similarly for $\varphi = -1$.

Correction The first formula holds for $0 < \operatorname{Re}\theta < \pi/2$, the second for $\operatorname{Re}\varphi > 0$ and $|\operatorname{Im}\varphi| < \pi/2$; for negative real parts the left sides must be replaced by $-\theta$ and $-\varphi$.

Lean `Carlson.TwoVariable.carlsonRC_cot_sq_csc_sq_complex`, `Carlson.TwoVariable.carlsonRC_coth_sq_csch_sq_complex` (`Carlson.TwoVariable.R.ElementaryValues`).

2.3 Theorem 7.4-2 (two-saddle asymptotics of Jacobi polynomials), p. 198

As printed A two-saddle approximation on $V = \{(x,y) : xy \neq 0,\ x^2 \neq y^2\}$, with uniform asymptotics defined by convergence of the quotient to one.

Problem At $\alpha = \beta = -\frac{1}{2}$, $(x,y) = (1,\mathrm{i})$, the polynomial vanishes in every odd degree (its normalization is nonsingular in every degree). No quotient of it by any sequence can converge to one along the odd degrees.

Correction Read the quotient with removable extensions, or state the result as a two-saddle expansion with an additive error controlled by the sum of the magnitudes of the two saddle terms. Neither formulation is refuted by the example; neither is yet proved here at general parameters.

Lean `Carlson.TwoVariable.carlson_polynomial_two_saddle_ratio_counterexample` (`Carlson.Jacobi.AsymptoticZeros`).

2.4 Table 9.3-1, row $2b = (1,3,3)$

As printed $C = (z-y)\,d/5$, with $d = (x-y)(y-z)(z-x)$.

Correction $C = (y-z)\,d/5$; the coefficients $C_G = 2(3x-t)$, $C_F = 3yz-s$, $C_A = x(t^2-2s-xt)$ are correct. Confirmed formally and numerically.

Lean `Carlson.carlsonR_table_9_3_1_row4` (`Carlson.Elliptic.ReductionTables`), stated with $C = -(x-y)(x-z)(y-z)^2/5$.

2.5 Exercise 9.7-2

As stated For $x, y, z, \lambda, \mu, \nu \ge 0$ with at most one of $x+\nu$, $y+\nu$, $z+\nu$ zero, $\lambda > \nu$, $\mu > \nu$, and ν given by Carlson's formula,

$$R_F(x+\lambda, y+\lambda, z+\lambda) + R_F(x+\mu, y+\mu, z+\mu) = R_F(x+\nu, y+\nu, z+\nu).$$

Counterexample $x = y = 1$, $z = 1/100$, with $\lambda, \mu > 0$ on the other branch of Euler's relation (9.7-11). The formula gives $\nu = 0$ and all stated hypotheses hold, but numerically the left side is ≈ 1.6794 while $R_F(x, y, z) \approx 1.4780$.

Correction Add the branch condition of Theorem 9.7-1, $(\lambda - \nu)(\mu - \nu) > f(\nu)$.

Lean `Carlson.carlsonRF_add_carlsonRF_shift` (`Carlson.Elliptic.ChapterNineExercises`).

2.6 Exercise 9.8-6

As stated For $a > 0$ and $Q(s) = as^4 + bs^3 + cs^2 + ds + e$ positive on $(0, \infty)$ (with $e \geq 0$), and $C(t) = t^3 + ft^2 + gt + h$ with

$$f = c + 6\sqrt{ae}, \qquad g = bd + 4c\sqrt{ae} + 8ae, \qquad h = \left(b\sqrt{e} + d\sqrt{a}\right)^2,$$

$$\int_0^\infty Q(s)^{-1/2}\,\mathrm{d}s = \int_0^\infty C(t)^{-1/2}\,\mathrm{d}t.$$

Counterexample $Q(s) = s^4 - 1.9s^3 + 2s^2 - 1.9s + 1$ is positive on $(0, \infty)$, but the two sides are $3.3721\ldots$ and $1.2574\ldots$. Since C depends on b, d only through bd and $(b\sqrt{e} + d\sqrt{a})^2$, it is unchanged when b and d change sign together, while the left side is not.

Correction Numerically the identity holds exactly when $b\sqrt{e} + d\sqrt{a} \geq 0$, which is when the substitution

$$t = 2as^2 + bs + 2\sqrt{a}\,\sqrt{Q(s)} - 2\sqrt{ae} \qquad (\text{for which } C(t) = t'(s)^2 Q(s))$$

is increasing. It is proved here for nonnegative coefficients; the sharp condition is not yet proved.

Lean `Carlson.integral_quartic_eq_integral_cubic_of_nonneg` (`Carlson.Elliptic.QuarticIntegral`).

References

[1] B. C. Carlson, *Special Functions of Applied Mathematics*, Academic Press, New York, 1977.